ETL MODERNIZATION ON GOOGLE CLOUD

Blueprint — internal reference

Legacy ETL → Google Cloud · ODI first adapter · assessment and migration

1 **Positioning and market**
GCP arm of Data Modernization, adapters, sectors, triggers, first reference case.
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SECTION 1

Positioning and Market

Where the asset sits, who buys it, what triggers the purchase, and the first reference case.

Positioning — GCP arm of Data Modernization p. 3

Market, triggers and first reference p. 4

Why now — the typical estate p. 5

Positioning — The Google Cloud Arm of Data Modernization

KOMTAS

An assessment that measures and prices, and a migration that delivers. ODI is the first source adapter; others follow the same pattern catalogue.

01 · POSITIONING

WHERE IT SITS IN THE PORTFOLIO

Data Modernization
delivery package: pattern decision (DWH / lakehouse / hybrid), then build

↓

ETL Modernization on Google Cloud
its GCP arm: legacy ETL estate → BigQuery, Dataform, Composer, Datastream

↓

- **A** — **ETL Migration Assessment**
  fixed price · 3–4 weeks
- **B** — **ETL Migration**
  waves · unit price or capped T&M

SOURCE ADAPTERS

ODI Oracle Data Integrator 11g/12c · first reference case	ready
Informatica PowerCenter target alternative: IDMC on Google Cloud (Komtaş licence line)	planned
SSIS SQL Server Integration Services	planned
Talend Talend Open Studio / Data Fabric	planned

No channel conflict. ODI is Oracle's product and is not resold by Komtaş, so moving an ODI estate does not compete with the Informatica licence business. When the source is PowerCenter, IDMC on Google Cloud is offered as the alternative target.

Market, Triggers and the First Reference

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Who buys it today, what makes them move, and where the first case stands.

01 · MARKET

TARGET SECTORS TODAY

- Retail
- Consumer
- E-commerce
- Media
- Manufacturing
- Holdings

Banks wait for a region

The largest ODI users in Türkiye are banks. BDDK-regulated workloads need a Google Cloud region in Türkiye — announced with Turkcell in November 2025, no opening date yet. When it opens, this asset is ready for that market.

TRIGGERS

Oracle licence renewal
DWH and ETL licences on the table

ODI skills gap
vendor-dependent operation

Hardware refresh
on-prem DWH end of life

Existing GCP commitment
consumption target to meet

Open point for the asset decision (S1)

Today there is one customer and no second concrete opportunity, which fails the repeatability gate. Sales and the Google partner manager are asked for the legacy-ETL-to-GCP pipeline; at least two concrete opportunities are needed. Candidate 08 (“work done with GCP”) may be the same thing and should be merged if so.

FIRST REFERENCE CASE

MACFit — ODI to Google Cloud

Status	proposal requested, Oct 2026
Estate	714 packages, 15 sources, Oracle DWH
Built by	third-party vendor
Email estimate	7–9 months, 4–5 people
Rev2 plan	508.8 pd · 3 DE ≈ 11.5 months
Pricing view	monthly T&M, not turnkey

Why Now — What a Legacy ETL Estate Usually Looks Like

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Findings that come up again and again when an ODI estate is inventoried, and what changes once it runs on Google Cloud.

01 · WHY NOW

Production only

No separate development or test environment; every change is made live.

No automatic alerting

Failures are found by a night operator or by users in the morning.

Knowledge with a few people

Often the vendor that built it; the owner cannot see all the logic.

No data quality checks

Duplicates, nulls and reconciliation are not checked systematically.

Hundreds of small jobs

Most jobs run in under a minute; the complexity is in the count.

Licence and skills cost

Oracle-based ETL and DWH licences; ODI skills are getting scarce.

Each of these is checked in the assessment and becomes a line in the migration plan.

WHAT CHANGES ON GOOGLE CLOUD

Dev, test and prod
separated, deployed by CI/CD

An alert on every failure
Cloud Monitoring, before users notice

Transformations as SQL
versioned, tested, with lineage (Dataform)

Ingestion as configuration
Datastream CDC instead of copy jobs

Pay for use
serverless BigQuery, no ETL server

SECTION 2

Method

Target platform, pattern catalogue, engine choice, the two services, and how the estimate is built.

Target platform and pattern catalogue — p. 7 – 8

Engine choice and ingestion as configuration — p. 9

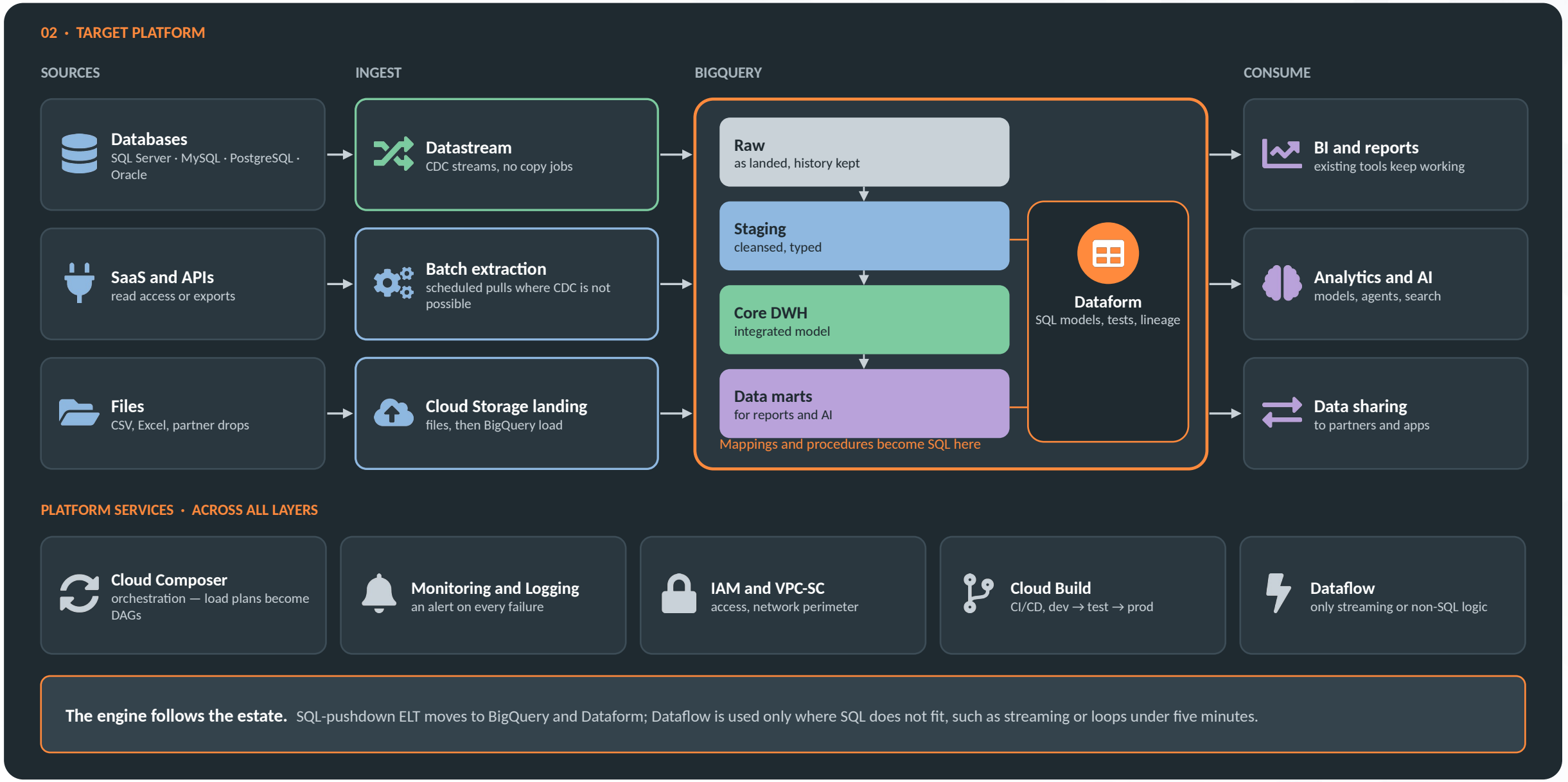
Two services · how the estimate is built — p. 10 – 11

Discovery toolkit · the 34% blind spot — p. 12 – 13

Target Platform on Google Cloud

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Sources land through CDC, batch extraction or files; transformations run as SQL in BigQuery; one orchestrator, one monitoring layer.



How ODI Maps to Google Cloud

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Every ODI object type has a target pattern. Each pattern is built once in the pilot and reused for every package of that type.

02 · PATTERN CATALOGUE

ODI OBJECT	GOOGLE CLOUD	HOW IT MOVES	
Mapping / Interface	Dataform SQLX model	SQL translated with BigQuery Migration Service, then reviewed	translate + review
Procedure (PL/SQL)	BigQuery scripting or stored procedure	logic rewritten, tested against the old output	rewrite
Package	Composer task group	steps and branches kept in the same order	template
Load Plan	Composer DAG	serial and parallel steps, retries, schedule	generated
Variable	Airflow or Dataform variable	refresh queries moved to the first task	template
Journalizing (CDC)	Datastream stream	no code; source CDC must be enabled	configuration
File LKM	Cloud Storage + BigQuery load	one landing pattern for all files	template
Knowledge Module	Generic landing and merge pattern	written once, reused by every package	template
Operator log	Cloud Logging + Monitoring	alerts, run history, dashboards	configuration

configuration lowest effort

template low

translate + review medium

rewrite highest

effort per package

Engine Choice and Ingestion as Configuration

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Two design decisions change the unit hours more than anything else; both are made in the assessment, before the price.

WHICH ENGINE FOR WHICH WORKLOAD

WORKLOAD IN ODI	TARGET	WHY
SQL-pushdown mapping (ELT)	BigQuery + Dataform	Same logic as SQL; tests, lineage and dependencies for free
PL/SQL procedure	BigQuery scripting	Rewritten; behaviour checked against the old output
Load plan / package	Cloud Composer	DAGs and task groups keep order, retries and schedule
1:1 copy from a database	Datastream	CDC configuration; no code if the source allows CDC
File load	Cloud Storage + load job	One landing pattern for all files
Streaming or < 5-minute loop	Dataflow	Only where SQL in BigQuery does not fit
Non-SQL logic (Java, Groovy)	Dataflow or Cloud Run	Case by case; counted as rewrite

Why not Dataflow for everything

The first email proposed Dataflow + BigQuery. For an estate of small SQL-pushdown jobs, Dataflow means rewriting SQL as pipeline code — more effort, more cost to run, harder to maintain. The argument to the account team is not “the number is wrong” but “the engine is wrong, and the number changes with it”.

Ingestion as configuration — the precondition

About 290 of 714 packages are 1:1 copies into ODS/EXT layers. With Datastream they become stream configuration plus one generic landing pattern. This needs CDC on the source databases; SaaS sources with read-only DB access fall back to batch extraction. It is the first question of the assessment.

Two Services — Measure First, Then Migrate

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The assessment turns an unknown estate into an inventory and a price. The migration is then run in waves against that inventory.

03 · TWO SERVICES

A — ETL Migration Assessment

Fixed price · 3–4 weeks

Focus	Measure the estate and price the migration
Evidence	Repository queries, discovery form, interviews
Access	Read-only to the ODI repository, or the script run by your team
Output	Inventory, complexity groups, target architecture, plan and price

B — ETL Migration on Google Cloud

In waves · duration set by the estate

Focus	Build the platform, convert, run in parallel, cut over
Pricing	Per package by complexity group, or time and materials with a cap
Team	Architect, data engineers, cloud engineer, project manager
Output	Running platform on Google Cloud, ODI switched off, runbooks handed over

Why not a fixed price straight away?

When an estate was built by a third party and is not fully documented, a fixed price either carries a large risk margin or breaks at the first surprise. The assessment removes that uncertainty first; the migration price then follows the measured inventory.

How the Estimate Is Built

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From the repository to a price in six steps. Unit hours are assumptions until the pilot calibrates them.

03 · ESTIMATE

1 Discovery script
reads the ODI repository: packages, mappings, procedures, load plans, runtime

2 Inventory
every package with sources, targets, schedule, run time and volume

3 Complexity groups
five structural groups, plus 1:1 ingestion handled as configuration

4 Unit hours × reserve
analysis, conversion and test hours per group; reserve for unknowns

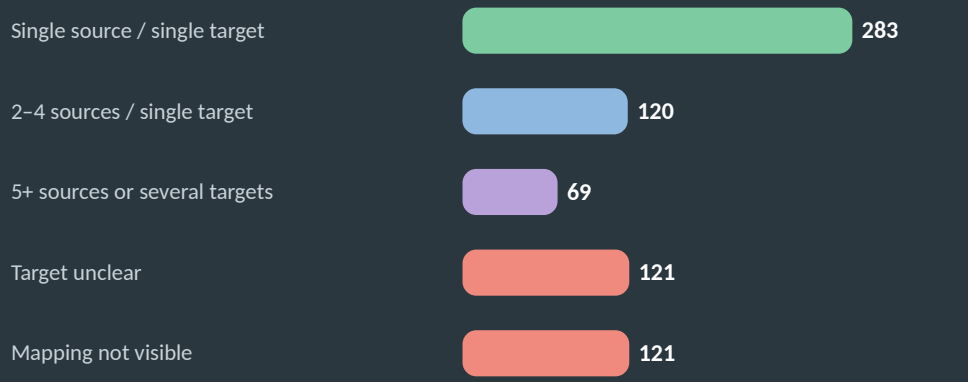
5 Pilot calibration
two critical flows end to end; unit hours corrected

6 Plan and price
team size against target date; per-package or capped T&M price

A recent retail estate

example · anonymised

714	15	236	daily
packages	source systems	source tables	all flows



242 packages (34%) need a closer look — usually procedure-heavy. The deeper discovery queries resolve them before the price is fixed.

About 290 jobs are 1:1 copies. If CDC can be enabled on the sources, they become Datastream configuration instead of code.

Discovery Toolkit — Form and Repository Queries

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The form collects what only people know; the queries read everything the repository already holds. Nothing is written to the repository.

DISCOVERY FORM · 13 SHEETS

- 0 Instructions
- 1 System architecture
- 2 Data inventory
- **3 Pipeline and job inventory ★**
- **4 ETL type split ★**
- **5 Performance and SLA ★**
- 6 Dependencies
- 7 Data quality
- 8 Error handling
- 9 Access and repository
- 10 Risks
- 11 Expected outputs
- 12 Sample flow

★ pre-filled by the queries; the customer adds business meaning only

REPOSITORY QUERIES · ODI 12c, READ-ONLY

#	QUERY	WHAT IT READS	FEEDS
00	Schema check	columns of every repository table used, before anything runs	—
01	Package inventory	steps, mappings, procedures, sources, targets, load plan	Sheet 3
02	Effort group	the five structural groups and 1:1 copy candidates	Effort model
03	Mapping complexity	joins, filters, expressions, aggregates, KMs	Unit hours
04	Procedures	lines, PL/SQL, dynamic SQL, DB links	Blind spot
05–07	Variables, CDC, files	refresh queries, journalizing, non-DB sources	Patterns
08	Load plans	structure, parallel steps, schedules	DAG design
09	Runtime	runs, min/avg/max minutes, success rate, errors	Sheet 5
10–11	Volumes	rows per day, table sizes on the target DWH	Sheet 2
12	Summary	counts per prefix, group and technology	Proposal

The 34% Blind Spot and the Deeper Discovery

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242 of 714 packages have no visible mapping or no clear target. They are counted at a medium rate until the queries below resolve them.

WHAT IS MISSING	QUERY THAT RESOLVES IT	WHAT IT CHANGES
Procedure logic	Procedure lines and stored text (SNP_TRT, SNP_LINE_TRT, text tables)	Rewrite vs template; PL/SQL size per package
Knowledge module per mapping	KM names from scenario and session tasks	Which steps are generic loads
Variables	SNP_VAR and refresh queries	Parameters moved to Composer / Dataform
Journalizing	Journalized datastores and models	Direct Datastream candidates
File loads	File technology in topology and steps	Cloud Storage landing pattern
Row counts and size	Target DB dictionary: num_rows, segments	Load windows, BigQuery cost estimate
Daily volume	Session task log: rows inserted / updated per day	Parallel run sizing, slot capacity

How it is used

The script (file 8) runs query 00 first to check repository columns for the exact ODI version, then the rest. The 81 points marked VERIFY in the script are confirmed on the first real repository and the script is updated. Unit hours are not changed from the queries alone; the pilot calibrates them.

SECTION 3

Komtaş Side

Effort model, reference estimate and pricing, plan and team, accelerators, quality gates, scope, package status.

Effort model · reference estimate and pricing — p. 15 – 16

Migration plan and team · accelerators — p. 17 – 18

Quality gates · scope and limits — p. 19 – 20

Package files and open items — p. 21

Effort Model — Groups, Unit Hours and Reserve

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Package counts come from the inventory; unit hours are starting values that the pilot corrects. Person-days = hours ÷ 8 × (1 + reserve).

UNIT HOURS PER PACKAGE · STARTING VALUES

EFFORT GROUP	ANALYSIS	CONVERSION	TEST	TOTAL
Single source / single target	0.5	2	0.5	3
2–4 sources / single target	1	3	1.5	5.5
5+ sources or several targets	2	4	2	8
Target unclear	1	3	1.5	5.5
Mapping not visible	1	3	1.5	5.5
1:1 ingestion — configuration	0.1	0.5	0.25	0.85

Grouping order: mapping not visible → target unclear → 5+ sources or several targets → 2–4 sources → single source. The 1:1 ingestion group applies only when CDC is possible on the source.

PHASES

PHASE	DAYS	WHO
Analysis and design	20	1 DE, cloud engineer, architect
Platform and pilot	20	2 DE, cloud engineer, architect
Conversion	effort ÷ DE	all DE; architect 20%
Cutover	20	2 DE, cloud engineer

Example · 714 packages

Base effort	**424 pd**
Reserve 20%	**84.8 pd**
Total	**508.8 pd**
3 DE	**≈ 11.5 months**
5 DE	**≈ 9 months**

If the ~290 copy jobs become Datastream configuration, the total falls by roughly 90 person-days — about one month with 3 data engineers.

The model ships as a workbook: paste the inventory, set the team size, read the plan and the Gantt.

Reference Estimate — Rev2 Against the First Email

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Same inventory, different assumptions. The table shows where the numbers come from and what moves them.

	EMAIL (12.10.2026)	REV2 PLAN	WITH DATASTREAM*
Basis	meetings, form header	714 packages, 5 groups	Rev2 + 1:1 copies as config
Data engineer effort	—	508.8 pd	≈ 420 pd
Data engineers	2–3	3	3
Total duration	7–9 months	230 days ≈ 11.5 months	≈ 10 months
DE needed for 9 months	—	5	4
Engine	Dataflow + BigQuery	not decided	BigQuery + Dataform + Datastream

* approximate: ~290 single-source copy jobs moved from 3 h to 0.85 h per package; all other assumptions as Rev2. Confirmed only after the CDC check and the pilot.

Pricing view

While the estate is uncertain, the migration is not offered turnkey. Two options: a unit price per package by effort group (person-days × day rate, reserve included), or monthly time and materials with a cap. The assessment is fixed price. Effort for the assessment itself is an initial SAT estimate of 25–35 person-days (architect, data engineer, part-time PM) and needs delivery confirmation.

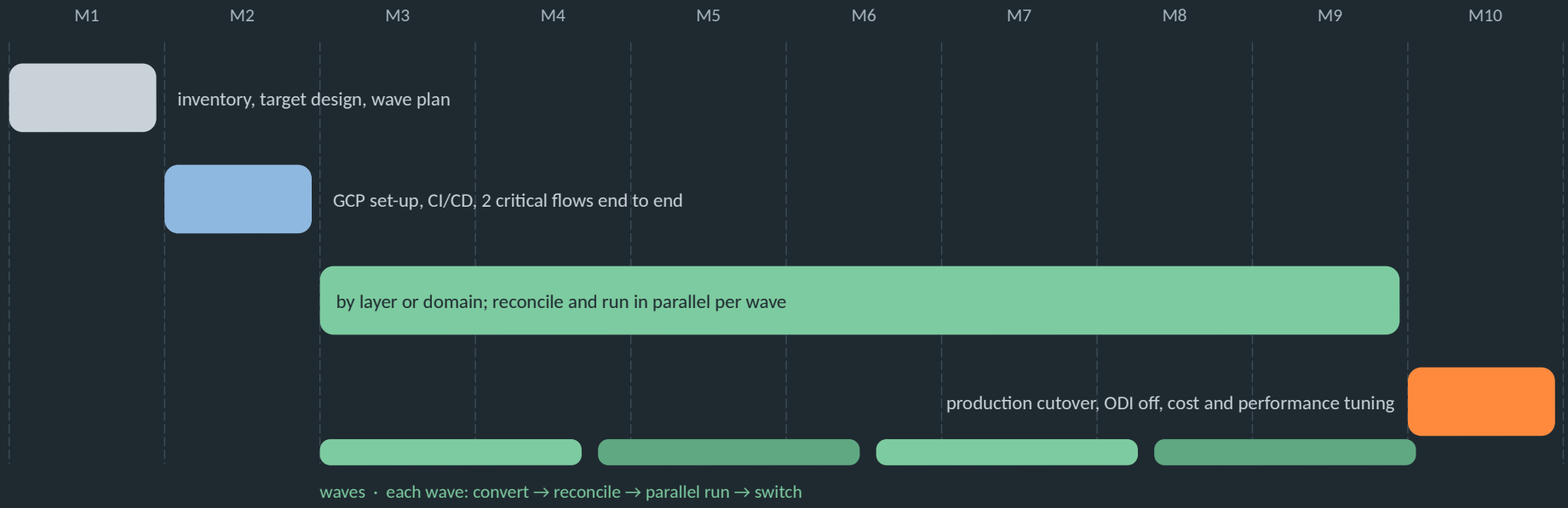
Migration Plan — Four Phases, Converted in Waves

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Phase 3 length is set by the inventory and the team size. Every wave is reconciled and run in parallel before it is switched over.

04 · MIGRATION PLAN

PHASES · INDICATIVE MONTHS



TEAM

- **Cloud Solution Architect** full in phases 1–2, about 20% after
- **Data Engineers** 2 to 5, set by the target date
- **Cloud Engineer / DevOps** full time
- **Project Manager** part time

QUALITY GATES

- **Reconciliation** row counts and checksums per table
- **Parallel run** old and new side by side
- **Cutover runbook** steps, owners, rollback

Accelerators and Funding

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Google Cloud services and programmes that shorten the migration or lower its cost. Eligibility and terms are set by Google per project.

BigQuery Migration Service	Batch SQL translation from Oracle and other dialects into BigQuery SQL; used for mapping and procedure SQL. The Migration API is offered at no charge.
Datastream	Serverless change data capture from MySQL, PostgreSQL, Oracle and SQL Server into BigQuery — replaces 1:1 copy jobs.
Dataform	SQL workflows in BigQuery with dependencies, assertions (tests), documentation and lineage; versioned in Git.
Cloud Composer	Managed Apache Airflow; ODI load plans and packages become DAGs and task groups.
Google Cloud RAMP	Partner-led migration programme with assessment support and funding or credits, subject to Google's eligibility review.

Product capabilities change; the assessment confirms current support for the exact source versions.

Reconciliation, Parallel Run and Cutover

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How each wave proves the new platform returns the same numbers before the old flow is switched off.

1 Reconcile

Every converted table is compared with its ODI counterpart after each load.

- Row counts per table and per load date
- Checksums on key and amount columns
- Aggregates on the measures reports use
- Differences logged with an owner and a cause

2 Run in parallel

Old and new run side by side on the same days.

- Typically two to four weeks per wave
- Daily reconciliation report to both teams
- Business owners confirm the critical reports
- Exit only when differences are explained

3 Cut over

A written runbook per wave, rehearsed once.

- Freeze, final load, switch of consumers
- Named owner for every step
- Rollback path kept until sign-off
- ODI flow disabled, then removed

Acceptance is a number, not an opinion. Each wave is accepted when reconciliation is within the agreed tolerance for the agreed period, and the business owner has signed the critical reports.

Scope and Limits

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What the services cover, what they leave out, and where they cannot be offered today.

IN SCOPE

- ODI packages, mappings, procedures, variables and load plans
- Ingestion from the existing sources into BigQuery
- Target model layers as they exist today (raw to marts)
- Orchestration, monitoring, CI/CD and access set-up
- Reconciliation, parallel run, cutover, ODI decommissioning
- Runbooks and handover to the operating team

OUT OF SCOPE

- Changes to the source systems themselves
- Redesign of the data model beyond agreed fixes
- Migration of BI reports and dashboards (separate scope)
- Licence costs of Google Cloud consumption
- Data that must stay on premises or in Türkiye by regulation

Regulated workloads

Workloads that must stay in Türkiye under BDDK or similar rules cannot move until a Google Cloud region in Türkiye is open. A region was announced with Turkcell in November 2025; its opening date is not yet set. Until then the services address retail, consumer, e-commerce, media, manufacturing and holding companies.

Package Files and Open Items

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DWH Assessment numbering with -en / -tr suffixes; A and B split as in Lakehouse. Files 3–6 follow once S1 is settled.

#	FILE	STATUS	NOTE
1	etl-migration-gcp-1-blueprint-en.pptx	ready	this document
2	etl-migration-gcp-2-musteri-sunumu-en.pptx	ready	20-minute main flow + appendix
3a / 3b	assessment / migration input forms	after S1	bucket model and staffing inputs
4a / 4b	assessment / migration proposal templates	after S1	fixed price vs unit price / capped T&M
5 / 6	customer / internal question sets	after S1	first question: CDC per source
7	etl-migration-gcp-7-discovery-formu-tr.xlsx	ready	generic DD form, 13 sheets
8	etl-migration-gcp-8-odi-repository-sorgulari-en.sql	draft	ODI 12c; 81 VERIFY points
9	etl-migration-gcp-9-efor-modeli-tr.xlsx	ready	groups × hours × reserve, plan, Gantt

OPEN ITEMS FROM THE 15-QUESTION ASSESSMENT

- S1 repeatability — one customer today; ask Sales and the Google partner manager for the pipeline (two concrete opportunities needed)
- S4 past effort records and S6 written acceptance criteria — Delivery
- S11 GCP / BigQuery / Dataform capacity — how many people besides Kadir
- S12 partner level and certified staff — Partner management
- Candidate 08 (work done with GCP) — check with Evren, merge if it is the same